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Elucidating governing factors of PFAS removal by polyamide membranes using machine learning and molecular simulations

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Per- and polyfluoroalkyl substances (PFASs) have recently garnered considerable concerns regarding their impacts on human and ecological health. Despite the important roles of polyamide membranes in remediating PFASs-contaminated water, the governing factors influencing PFAS transport across these membranes remain elusive. In this study, we investigate PFAS rejection by polyamide membranes using two machine learning (ML) models, namely XGBoost and multimodal transformer models. Utilizing the Shapley additive explanation method for XGBoost model interpretation unveils the impacts of both PFAS characteristics and membrane properties on model predictions. The examination of the impacts of chemical structure involves interpreting the multimodal transformer model incorporated with simplified molecular input line entry system strings through heat maps, providing a visual representation of the attention score assigned to each atom of PFAS molecules. Both ML interpretation methods highlight the dominance of electrostatic interaction in governing PFAS transport across polyamide membranes. The roles of functional groups in altering PFAS transport across membranes are further revealed by molecular simulations. The combination of ML with computer simulations not only advances our knowledge of PFAS removal by polyamide membranes, but also provides an innovative approach to facilitate data-driven feature selection for the development of high-performance membranes with improved PFAS removal efficiency.

Per- and polyfluoroalkyl substances (PFASs) are anthropogenic compounds that possess a fluorinated carbon backbone and varying functional head groups. The strong carbon-fluorine bonds and helical conformation that shields the redox-reactive carbons on the inside[1] lead to exceptional stability of PFASs[2,3] with high resistance to water, oil, and heat[4]. These unique properties make PFASs suitable for a broad range of commercial applications, such as nonstick cookware, stain-resistance products, and flame retardants[4–7], but the persistence, toxicity, and bioaccumulative potential of PFASs have garnered considerable concerns regarding their impacts on both human and

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ecological health[8,9]. The health concerns on long-chain PFASs, such as perfluorooctane sulfonic acid (PFOS) and perfluorooctanoic acid (PFOA), have led to gradual phase-out in long-chain PFAS production within North America and Europe[10], which is accompanied by a shift towards higher productions of shorter-chain alternatives[11–13]. However, recent investigations also reveal toxicity associated with short-chain PFAS compounds due to their similar persistence and higher mobility compared to their long-chain counterparts[11,12].

The ubiquitous presence of PFASs in global water resources has raised alarm as reported by numerous studies[4,14–18]. It has been suggested that PFAS-contaminated drinking water is one of the major pathways of PFAS exposure to humans[19–21], resulting from ineffective PFAS removal by conventional water treatment processes[19,22]. Recently, the U.S. Environmental Protection Agency (USEPA) proposed the final National Primary Drinking Water Regulation for six PFAS species, setting extremely low maximum contamination levels (MCLs) for PFOA and PFOS at 4 ng L^-1 as well as other PFASs including perfluorohexane sulfonic acid (PFHxS), perfluorononanoic acid (PFNA), and GenX at 10 ng L^-1[23]. Such stringent regulations, along with the threat of an expanding list of PFASs with regulatory implications, warrant the need to develop effective technologies for PFAS removal[23]. Such technologies are vital for the potential upgrade of water treatment facilities to meet the corresponding regulations.

Membrane separation represents an important approach of treating PFAS-contaminated water sources[24–26]. Previous studies have evaluated the efficiencies of PFAS removal by nanofiltration (NF) and reverse osmosis (RO), in which a dense polyamide membrane with sub-nanometer pores is used to allow faster and selective transport of water over PFAS[26–30]. While several studies have applied computational chemistry techniques to predict PFAS degradation and adsorption[31–33], to the best of our knowledge, no effective approaches for predicting membrane separation performance in PFAS removal, which is essential for rational membrane selection and design, have been reported in the literature. Such a knowledge gap is largely due to the lack of a mechanistic understanding of PFAS interactions with polyamide membranes at both the water-membrane interface and the molecular level. Despite the importance of membrane and PFAS properties in governing PFAS transport, a fundamental property-structure-performance relationship has not been established for PFAS removal by polyamide NF and RO membranes. Various PFAS compounds, which are well beyond the well-studied PFOS and PFOA, may vary in functional group substitution, ionization state, chain length, and other characteristics[34,35]. The diverse properties of PFASs, coupled with the existence of membranes with varied perm-selectivity, make the identification of key factors governing PFAS removal in membrane treatment a particularly complex and challenging task. Predictive models that can accurately estimate the efficiency of PFAS removal while unveiling the influences of membrane and molecular properties on PFAS transport across polyamide membranes are critical. Such models not only contribute to the advancement of knowledge but also facilitate wide adoption of polyamide NF and RO membranes for the remediation of PFAS-contaminated waters.

In recent years, machine learning (ML) algorithms have been increasingly applied to predicting membrane performance and investigating the governing mechanisms of membrane separation[36–40]. Thanks to its powerful capability of leveraging large datasets, ML can uncover intricate relationships among multiple variables. While ML has been successfully used to forecast the efficiency of polyamide NF and RO membranes in removing organic micropollutants[36,41,42], its applications to PFAS removal are very limited[40]. Furthermore, ML models predicting membrane performance in the literature have predominantly relied on molecular descriptors, such as molecular weight and logarithm of n-octanol/water partition coefficient (log K_{ow}), while direct utilization of chemical structures as input variables has been rarely explored. The incorporation of chemical structure in ML models can provide a more in-depth understanding of how specific structural features influence the behavior of PFAS transport during membrane filtration. However, integrating information of chemical structure into ML models remains a significant challenge given the diverse molecular configurations of PFAS compounds.

Transformer neural network, initially developed for natural language processing (NLP), has been successfully applied to computer vision and material design[43–45]. Unlike traditional neural networks, transformers apply a novel mechanism known as “attention”, which allows them to selectively prioritize input variables during model training[46]. This distinct mechanism enables transformers to effectively analyze and identify the interdependencies between variables, highlighting the influences of individual elements within a sequence. By utilizing simplified molecular input line entry system (SMILES) strings to represent the intricate chemical structures of PFAS compounds, the robust capabilities of transformers in analyzing sequential data offer a promising approach to exploring the impacts of chemical structures on the transport of PFASs across polyamide membranes.

In this study, we employ XGBoost and multimodal transformer models to elucidate the performance and underlying mechanisms of PFAS removal by polyamide NF and RO membranes (Fig. 1). Using training data collected from existing literature sources, the predictive performances of the ML models are evaluated with our experimental data obtained from NF treatment of aqueous film forming foam (AFFF)-contaminated feedwater. Given that AFFF contains a diverse set of PFAS compounds, evaluating the ML models with this PFAS mixture serves as a robust method to validate the accuracy and generalizability of models in predicting membrane performance for PFAS removal. To investigate the PFAS transport mechanisms with XGBoost model, we employ Shapley additive explanation (SHAP), a model interpretation method rooted in the cooperative game theory[47], to analyze the key governing factors influencing PFAS removal by polyamide membranes. We then introduce a novel methodology utilizing a multimodal transformer model to investigate the role of chemical structure in governing PFAS removal through heat maps, which visualize the contributions of individual chemical elements to the model prediction. The integrated interpretation applied to XGBoost and multimodal transformer models provides macroscopic insights related to membrane and PFAS properties, as well as microscopic perspectives pertaining to chemical structure. This approach enhances our understanding of PFAS removal by offering a comprehensive analysis at both levels. The molecular mechanisms underlying PFAS removal by polyamide NF and RO membranes unveiled by the ML models were further probed by molecular dynamics (MD) simulations. We conclude with a discussion on the implications of our findings to the use of NF and RO as competitive and vital technologies to achieve efficient remediation of PFAS-contaminated waters.

Results and discussion

Prediction performance of ML models

The XGBoost and multimodal transformer models were trained with 457 data points collected from existing literature sources, which include PFAS removal pertaining to 21 PFAS species and 16 types of polyamide NF and RO membranes. The model performance was subsequently assessed using 77 data points obtained from our own experiments. By evaluating the model accuracy with a complex mixture of PFASs from AFFF-contaminated feedwater (Supplementary Data 1), we demonstrated the performance of the ML models under a more challenging condition (than using individual PFAS). The predictive accuracy was examined using root mean squared error (RMSE) and mean absolute error (MAE). As shown in Fig. 2, the XGBoost model has a RMSE of 8.4% and a MAE of 6.4%, while RMSE and MAE of the multimodal transformer model are 5.9% and 4.5%, respectively (Supplementary Table 1). Although the multimodal transformer model shows slightly better predictive accuracy, such minor differences

suggest that these models are comparable in their predictive performance. Compared to previous studies using ML to estimate membrane performance (RMSE of 4.6–17.5%; MAE of 6.1–13.2%)[36–38,41,48], the XGBoost and multimodal transformer models in this study exhibited comparable or superior predictive accuracy. Expanding the dataset to include a broader range of PFAS removal rates and standardizing experimental protocols across studies are crucial to enhance model accuracy and reliability further. Additionally, as shown by our recent publications[36,37], ML models with strong predictive performance can adequately capture solute transport mechanisms. Accordingly, we evaluate the potential of our ML models to reveal the governing factors of PFAS removal by polyamide NF and RO membranes in the next sections.

Identifying key factors for PFAS removal by the SHAP method

Although ML techniques have been increasingly applied in the fields of environmental science and engineering[36–40,49], the black box nature of ML makes it challenging to understand the decision-making process. Explainable artificial intelligence (XAI) has been used to explore the contributions of input variables and the underlying knowledge of ML models, enhancing the trust and confidence of the model predictions[36,37,50–53]. In this section, we implemented the SHAP method, a type of XAI based on the cooperative game theory, to elucidate the knowledge of the XGBoost model regarding the governing mechanisms of PFAS removal by polyamide membranes.

As shown in Fig. 3a, the SHAP summary plot illustrates the influence of each input variable on the model prediction. The y-axis

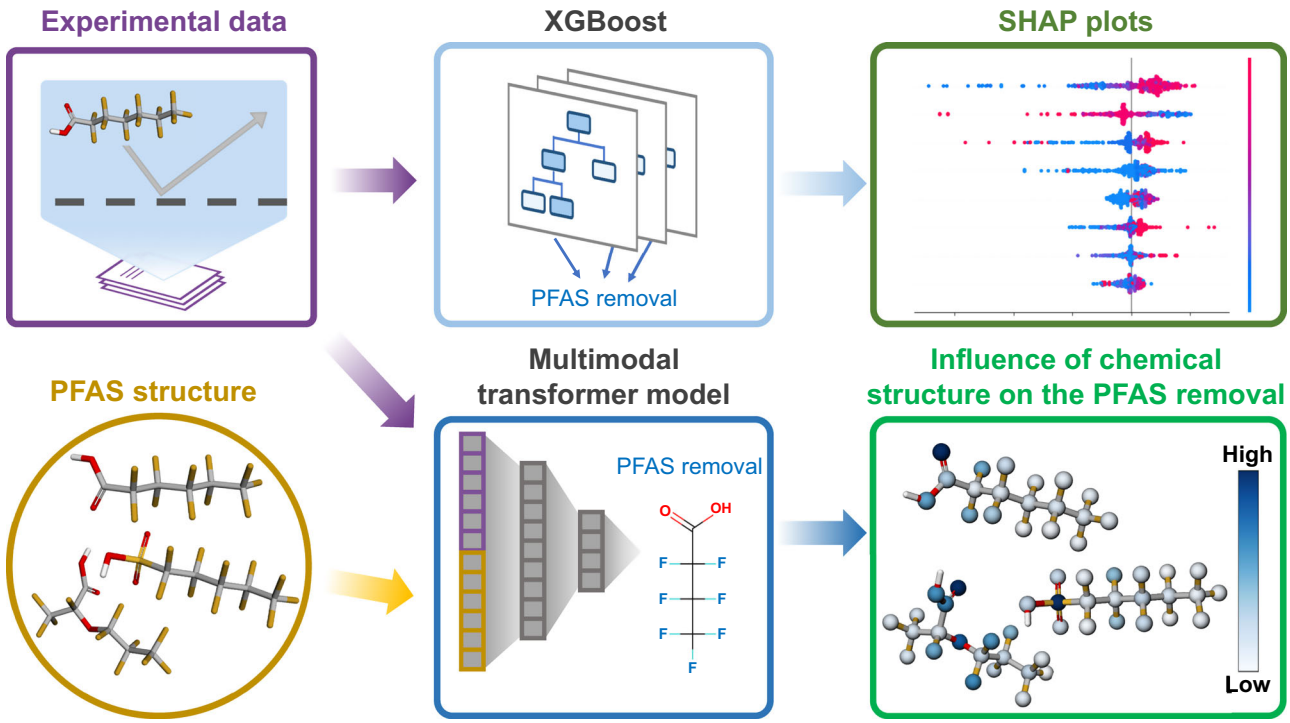


Fig. 1 | Workflow of XGBoost and multimodal transformer models for predicting PFAS removal rates by polyamide NF and RO membranes. The SHAP method was used for the interpretation of the XGBoost model, while attention scores of layers from the multimodal transformer model were extracted for presenting the importance of each PFAS atom to the model prediction.

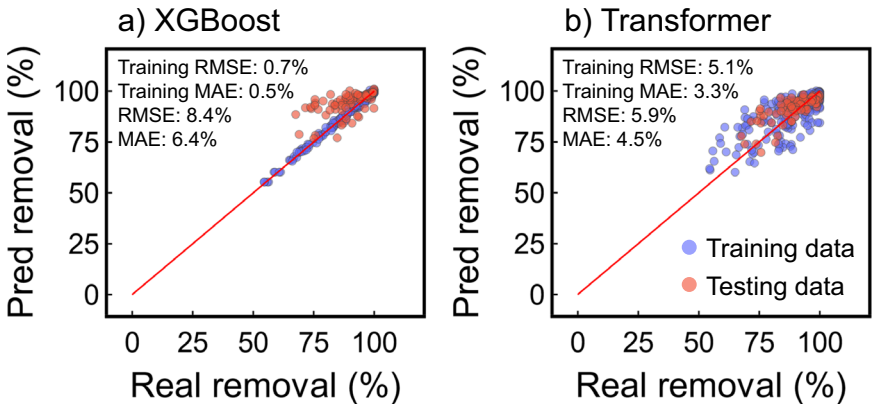


Fig. 2 | Predictive performance of XGBoost and the multimodal transformer models for PFAS removal. For the XGBoost model, numerical variables representing PFAS characteristics, membrane properties, and operational conditions were used as input variables. The multimodal transformer model used the same numerical variables in addition to SMILES strings that encode chemical structure of PFAS molecules. The x- and y-axes present the real and predicted removal of PFASs, respectively. Source data are provided as a Source Data file.

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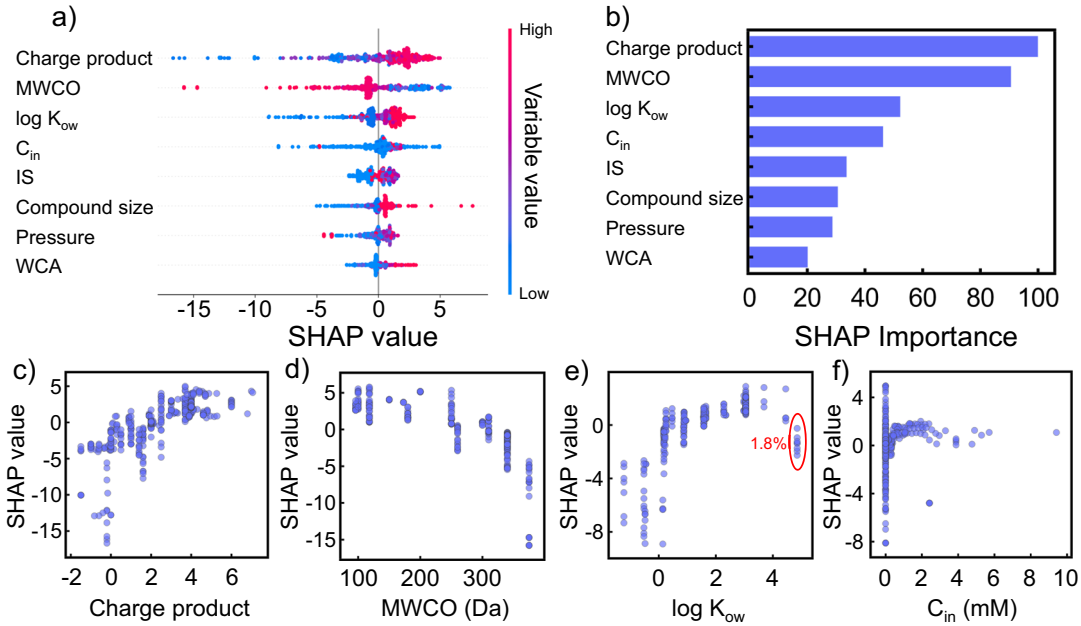


Fig. 3 | SHAP analysis of key variable influencing PFAS removal prediction by the XGBoost model. a SHAP summary, **b** SHAP importance, and SHAP dependence plots for (**c**) charge product, (**d**), molecular weight cut-off (MWCO), (**e**) log K_{ow}, and (**f**) initial concentration of PFAS (C_{in}) obtained from the XGBoost model. IS and WCA refer to ionic strength and water contact angle of membranes, respectively. For easier comparison, each importance score is normalized by the importance score of the input variable with the highest importance in Fig. 3b. The encircled data points in Fig. 3e indicate outliers. Source data are provided as a Source Data file.

arrangement indicates the hierarchical importance of the input variable, with importance descending from the top to the bottom. The colors of dots, combined with their SHAP values, demonstrate the correlations of input variables with their SHAP values. To further quantify these SHAP values, we calculated the average of absolute SHAP values of all data points (i.e., the SHAP importance score). As presented in Fig. 3b, the input variables with the largest contributions to the model predictions (i.e., the highest SHAP importance scores) are charge product (whose definition is described in the Methods section), molecular weight cut-off (MWCO), log K_{ow}, and initial concentration of PFAS (C_{in}). Considering the membrane separation mechanisms related to those input variables, electrostatic interaction, which is directly represented by the charge product, is the most important separation mechanism that determines the model predictions of PFAS removal by polyamide NF and RO membranes, followed by size exclusion (manifested by MWCO) and adsorption (manifested by log K_{ow} and C_{in}).

Although the SHAP importance score reveals the dominant factors that possess the highest weights to determine the model predictions, the correlation of each input variable with the model output, which is essential to understanding whether the knowledge learned by ML models aligns with the domain knowledge, is still unclear. Hence, we further probed the impacts of the four top-ranked input variables (i.e., charge product, MWCO, log K_{ow}, and C_{in}) using SHAP dependence plots (Fig. 3c–f), which elucidate how the SHAP value changes as a function of each variable. As shown in Fig. 3c, the SHAP values increase with charge product (i.e., an increase of charge product leads to an increase of PFAS removal rate). Due to the acidic functional groups (e.g., carboxyl and sulfonate groups) and strong electronegativity of fluorine[54], most PFASs exhibit low pK_a values (typically $pK_a < 1$, Supplementary Table 2). Therefore, PFASs are subject to nearly complete deprotonation and carry negative charges under most experimental pH conditions (pH of 3.5–10.0) from which the PFAS removal data were collected. Except for perfluorooctane sulfonamide (FOSA), electrostatic repulsion (i.e., charge product>0) exists between the PFASs and the membrane surface, due to the negative charge of polyamide membranes as a result of carboxyl groups generated from hydrolysis of the unreacted acyl chloride (−C(=O)Cl) groups[55]. As electrostatic repulsion between the negatively charged PFASs and membrane surface gets stronger (an increase of charge product), a higher removal rate of PFAS is correctly predicted by the XGBoost model (higher SHAP values).

The contribution of MWCO to the model prediction is presented in Fig. 3d. The decrease of MWCO leads to an increase of SHAP values, indicating that lower PFAS removal rates are predicted by the ML model for membranes with higher MWCO values, which are typically related to larger membrane pore sizes. This result is consistent with the mechanism of size exclusion for membrane separation. For example, the data points with low SHAP values (e.g., less than −4.2, indicating low PFAS removal) are associated with membranes with the highest MWCO (e.g., the XN45 membrane), which are known as loose polyamide NF membranes that have limited solute-retaining capability.

The third- and fourth-ranked variables are log K_{ow} and the initial concentration of PFAS (C_{in}), both of which are related to adsorption (Fig. 3e, f). Figure 3e shows the influence of log K_{ow} on the model predictions. Except for a small fraction (1.8%) of outliers, higher SHAP values were generally observed as a result of an increase of log K_{ow}, which corresponds to a higher tendency of adsorption[56,57]. As reported in the literature[56,58,59], adsorption of PFASs to the membrane surface contributes to their removal by NF membranes, especially at the early stage of membrane filtration. Also, the binding kinetics for long-chain PFASs are faster than short-chain PFASs due to their stronger hydrophobic interactions with membranes[24,56]. While log K_{ow} primarily represents the hydrophobicity of compounds[60,61], it is also an indirect indicator of PFAS size that is positively correlated to PFAS removal rate based on size exclusion. As the chain length of PFAS species increases, its hydrophobic nature typically gets stronger[59,62], leading to a higher log K_{ow}. This intertwined relationship between molecular size and hydrophobicity could be attributed when explaining the correlation between SHAP value and log K_{ow}.

Despite a high contribution of C_{in} to the model predictions, we did not observe a clear relationship between C_{in} and SHAP values (Fig. 3f).

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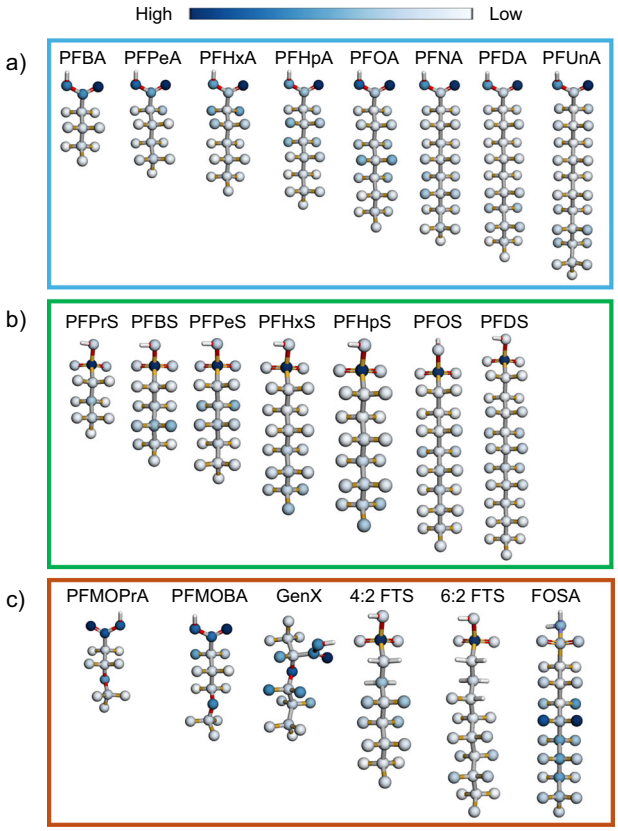


Fig. 4 | Heat maps highlighting attention scores of atoms for PFASs. Heat maps categorize PFAS into (**a**) PFCAs (blue box), (**b**) PFSAs (green box), and (**c**) other PFASs (orange box). The details of those PFASs categories are shown in Supplementary Table 2. The chemical structures of the PFAS compounds shown in this figure are also presented in Supplementary Fig. 2. Source data are provided as a Source Data file.

This lack of trend might be due to a broad range of C_{in} used in the literature for our dataset. Although a majority (~70%) of the data points are associated with low PFAS concentrations of less than 20 μM, ~30% of data points involve higher C_{in} up to higher than 9 mM (Supplementary Fig. 1). Given such a skewed data distribution, it is challenging for the ML model to capture a clear, linear correlation of C_{in} with the SHAP value across the entire range. Even though, the XGBoost model is still able to capture the impact of C_{in} from several small portions of the data range without compromising the model performance due to the nature of gradient boosting tree.

Probing key factors for PFAS removal by transformer model

Although the SHAP approach provides essential insights into the key factors that govern PFAS removal by polyamide NF and RO membranes, it remains elusive how chemical features of PFASs influence their transport across polyamide membranes. To bridge this knowledge gap, we implemented multimodal transformer model, which is known for its effectiveness of processing sequential data like SMILES strings. The heat-map visualization of the attention score of each atom from the multimodal transformer model enables us to identify specific atoms (and the corresponding functional groups) that have the highest importance in model predictions. Based on the functionality of PFAS species, the heat maps were grouped into perfluoroalkyl carboxylic acids (PFCAs), perfluoroalkyl sulfonic acids (PFSAs), and other PFASs (Fig. 4), where the colors of the atoms depict the values of attention scores.

As shown by the heat maps of PFCAs (Fig. 4a), the oxygen atoms of the carbonyl group (as part of the carboxyl group) exhibit the highest attention scores across all the PFCA compounds, regardless of the chain length. Similarly, the sulfur atom within the sulfonate group exhibited the highest attention scores for PFSA compounds (Fig. 4b). The particular functional groups associated with those atoms displaying the highest attention scores deprotonate at the experimental pH and result in a negative charge, which largely governs the electrostatic interactions of PFASs with the membrane surface. In contrast, carbon and fluorine atoms tend to exhibit lower attention scores. Therefore, in alignment with the insights revealed by the SHAP method (Fig. 3), the multimodal transformer model consistently identifies electrostatic interaction as the predominant mechanism influencing the model prediction of polyamide NF and RO membrane performance.

The heat maps for other PFAS compounds than PFCAs and PFSAs, including perfluoro-3-methoxypropanoic acid (PFMOPrA), perfluoro-4-methoxybutanoic acid (PFMOBA), 2,3,3,3-tetrafluoro-2-(heptafluoropropoxy)propanoic acid (GenX), 1H,1H,2H,2H-perfluorohexane sulfonic acid (4:2 FTS), 1H,1H,2H,2H-perfluorooctane sulfonic acid (6:2 FTS), and FOSA, are shown in Fig. 4c. For PFMOPrA, PFMOBA, and GenX, the oxygen atoms associated with the carbonyl group exhibit the highest attention scores, suggesting that the dominant role of electrostatic interaction in regulating PFAS removal also applies to those carboxyl-containing PFAS species. Interestingly, the oxygen atoms representing the ether (−O−) group also show high attention scores for these PFAS species. The presence of an ether group increases the hydrophilicity of PFASs and thus alters their interactions with the membrane (the role of ether group in PFAS-membrane interactions is further discussed in the next section). Also, the heat maps of 4:2 FTS and 6:2 FTS, both of which contain a sulfonate group, demonstrate a similar trend to that of PFSAs. In these cases, the sulfur atom shows the strongest influence on the model predictions. However, the highest attention score of FOSA is attributed to the fluorine atoms at the center of the compound. Unlike other PFAS compounds investigated in this study, the pK_a value of FOSA is close to neutral pH (Supplementary Table 2) due to the presence of sulfonamide group. This leads to a reduced degree of deprotonation and consequently, much weaker electrostatic interaction between FOSA and membrane surface. Thus, multimodal transformer model adeptly identifies functional distinctions between FOSA and other PFAS compounds, uniquely prioritizing the estimation of compound size. This estimation, indicated by the distance between the compound's center and head group, emerges as a pivotal chemical feature for predicting FOSA removal.

We further quantify the influence of chemical structure on PFAS removal by calculating the relative percentage of attention score (out of the total attention score summed for all the atoms) for each functional group (Supplementary Fig. 3). Compared to the carboxyl group of PFCAs with the same chain length (excluding the carbon in the functional group), the sulfonate group of PFSAs generally has a higher attention score percentage. Although both PFSAs and PFCAs are easily deprotonated, the pK_a values of PFSAs are lower than those of PFCAs (Supplementary Table 2), suggesting more negative molecular charge and consequently stronger electrostatic interactions with membrane surfaces for PFSAs. Such an effect is further supported by the atomic charge (i.e., the distribution of electron density across individual atoms in a molecule[63]) of PFCAs and PFSAs (Supplementary Figs. 4 and 5). Compared to the oxygen atoms of carboxyl groups, the oxygen atoms of sulfonate groups exhibit more negative atomic charges, indicating stronger electrostatic interactions with membrane surfaces. Thus, the multimodal transformer model discerned such differences in PFAS-membrane interactions when making predictions of PFAS removal by polyamide NF and RO membranes. Additionally, for other PFAS species than PFCAs and PFSAs, the functional groups pertaining to the atom where the atomic charge is the most negative generally match those with the highest attention scores from the

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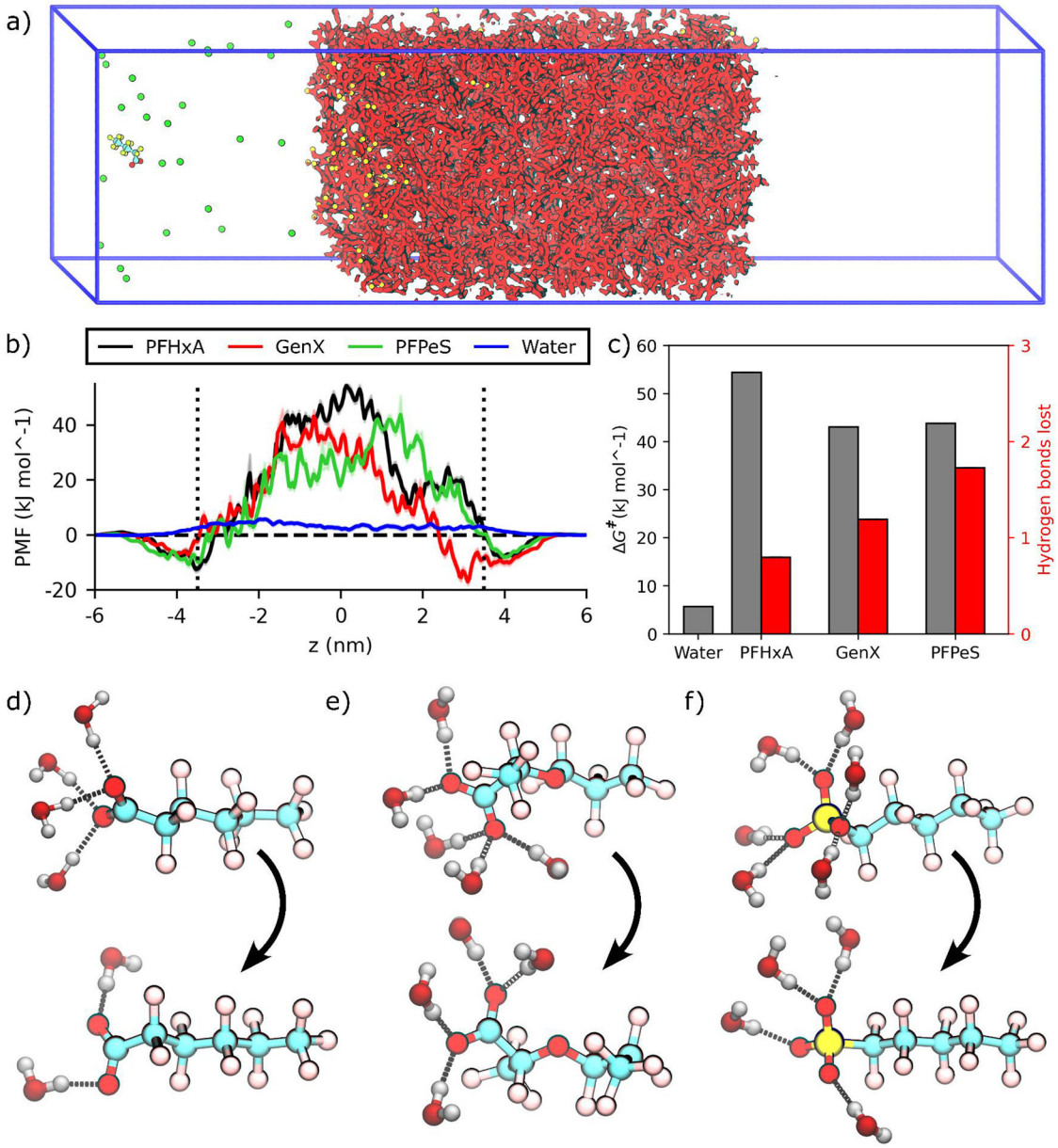


Fig. 5 | Molecular dynamics simulation of PFAS rejection mechanisms in a charged PIP-TMC membrane. a The initial system configuration of PFAS (cyan: carbon, yellow: hydrogen, and red: oxygen), polyamide membrane (red; yellow: deprotonation sites), and neutralizing sodium ions (green). The water molecules are not shown. **b** the potential of mean force for moving PFAS across the polyamide membrane. The dashed line marks the reference free energy. The dotted line marks the boundaries of the dense layer of the polyamide membrane. **c** Comparison of the effective free energy barrier in the PMF profile (gray) and the number of hydrogen bonds lost by the PFAS at the dense layer of the polyamide membrane (red). Examples of hydrogen bonding (black dashed line) of (**d**) PFHxA, (**e**) GenX, and (**f**) PFPeS in bulk water (top) and the dense layer of the polyamide membrane (bottom). Cyan: carbon, pink: fluorine, white: hydrogen, red: oxygen, and yellow: sulfur. Source data are provided as a Source Data file.

multimodal transformer model (except for FOSA), as shown in Fig. 4 and Supplementary Fig. 6.

Therefore, both ML models (i.e., the XGBoost model and multimodal transformer model) reveal unequivocally that functional groups relating to electrostatic interactions are the major structural features that govern the removal of most PFASs by polyamide NF and RO membranes. The roles of functional groups in regulating PFAS transport across polyamide membrane will be further explored by computer simulations in the next section.

Mechanisms of PFAS transport revealed by MD simulations

MD simulations were performed to probe the molecular mechanisms of PFAS transport across a charged and crosslinked piperazine (PIP)-trimesoyl chloride (TMC) polyamide membrane (Fig. 5a). We selected three PFASs including PFHxA, GenX, and PFPeS that contain different functional groups (carboxyl, carboxyl+ether, and sulfonic) with identical carbon chain lengths. Due to their low pK_a values (Supplementary Table 2), ionized PFAS molecules were used in our simulations. The potential of mean force (PMF) profiles (Fig. 5b) were obtained by umbrella sampling simulations followed by the weighted histogram analysis method (WHAM)[64] to probe the differences between the PFAS transport. Additionally, the PMF of transporting a water molecule was calculated for comparison. Initially, the PMF decreases as PFAS moves from bulk water towards the polyamide membrane ($|z|$ from 5 to 3.5 nm), denoting spontaneous adsorption due to their hydrophobic character (e.g., Liu et al.[65]). The adsorption free energy is observed to

be −12 to −9 kJ mol^-1. As the PFAS enters the dense layer of the polyamide membrane (|z|<3.5 nm; dotted lines in Fig. 5b) and moves towards the center, the PMF increases, denoting exclusion free energy arising from size, charge, and dielectric exclusion effects. The exclusion free energy defined as the difference between maximum and minimum PMF is in the range of 55-65 kJ mol^-1. Both the adsorption and exclusion free energy contribute to PFAS rejection. The adsorption free energy presents as an energy barrier for PFAS to move from the polyamide membrane to permeate-side bulk water, whereas exclusion free energies prevent PFAS from entering the membrane. In comparison to PFAS, water does not show any adsorption and has an exclusion free energy of 5.7 kJ mol^-1. The higher energy barriers of PFAS molecules compared to water indicates that PFAS transport is significantly hindered, while water molecules can easily permeate through the membrane. Since the exclusion free energy of PFAS is much higher than its adsorption free energy, the former is more significant for PFAS rejection. These results are consistent with ML results that find charge product and MWCO is more significant for PFAS rejection than log K_{ow}.

The permeability and rejection trends of PFAS are dependent on the maximum PMF that reflects the energy barrier $\Delta G^{\ddagger}$ associated with a PFAS reaching the center of the polyamide membrane. The $\Delta G^{\ddagger}$ of PFAS is much higher than water and follows the order GenX ≤ PFPeS <PFHxA (Fig. 5c). To interpret the order of $\Delta G^{\ddagger}$, van der Waals and electrostatic energies (Supplementary Fig. 7), total number of hydrogen bonds (Supplementary Fig. 8), average orientation of PFAS (Supplementary Fig. 9), and root-mean-squared-deviation of PFAS (a measure of conformational change; Supplementary Fig. 10) were calculated along the reaction coordinate. The increase in electrostatic energy as PFAS moves from bulk water toward the center of the membrane (Supplementary Fig. 7) follows the same order as that of $\Delta G^{\ddagger}$ (GenX <PFPeS <PFHxA). The change in the van der Waals (modeled using Lennard-Jones interactions) and electrostatic (modeled using Coulombic interactions) energies along the reaction coordinates (Supplementary Fig. 7) demonstrates that change in electrostatic interactions is more dominant than van der Waals. Since the PFAS molecules have the same net charge, the difference in electrostatic energies indicates the influence of dipole-dipole and dipole-charge interactions between PFAS and polyamide membrane. This implies that electrostatic attraction from dipole-dipole and dipole-charge interactions between PFAS and the polyamide membrane lowers the energy barrier created by electrostatic repulsion. While the input variables of ML models do not contain dipole properties of PFAS, higher attention scores in carboxyl, sulfonate, and ether groups indicate that the importance of electronegative atoms, which play a significant role in dipole interactions, are captured by the models to a certain degree. The order of van der Waal's interaction energy is opposite to that of $\Delta G^{\ddagger}$ and much lower in magnitude than electrostatic energy. The fluctuations in the number of hydrogen bonds of PFAS along the reaction coordinate z negatively correlated with the fluctuations in the PMF (Supplementary Fig. 8), indicating that dehydration of PFAS is likely to play a role in regulating PFAS rejection. The total number of hydrogen bonds lost by PFAS as it moves from bulk water (|z|> 5 nm) to the center of the membrane (|z|<3.5 nm) is shown in Figs. 5c–f. The number of hydrogen bonds lost is indicative of PFAS dehydration and follows the order of PFHxA <GenX <PFPeS. Such an order does not follow the order of $\Delta G^{\ddagger}$ but matches that of log K_{ow} of PFAS (Supplementary Table 2) because higher lipophilicity leads to higher propensity of PFAS dehydration. Therefore, while dehydration of PFAS might influence the membrane rejection performance, it is not as influential as the effects of electrostatic and size exclusion. This result is also consistent with ML results where log K_{ow} was found to be less influential than parameters associated with electrostatic and size exclusion (e.g., charge product and MWCO). The PFAS molecules exhibited significant changes in orientation and conformation near the center of the membrane, which did not correspond to fluctuations in the PMF (Supplementary Fig. 9). We attribute these changes to the tortuous nature of the pore pathways in the polyamide membrane.

Environmental implications

In this study, we employed XGBoost and multimodal transformer models to elucidate the governing factors of PFAS removal by polyamide NF and RO membranes. Both XGBoost and multimodal transformer models exhibited high prediction accuracy (with RMSE and MAE values of 6–8% and 5-6%, respectively), effectively correlating multiple variables (including membrane and PFAS properties as well as operational conditions) with membrane efficiency of PFAS rejection. The XGBoost model, interpreted with the SHAP method, and the multimodal transformer model, incorporating SMILES strings, consistently highlight the importance of electrostatic interaction in governing PFAS removal during NF and RO processes. MD simulations results agree well with ML results in identifying electrostatic interaction as a major governing factor for PFAS removal by polyamide membrane. Also, MD simulations confirm the importance of functional groups in regulating PFAS transport. Specifically, MD simulations demonstrate that electrostatic attraction from dipole interactions reduces the electrostatic exclusion effects. Additionally, PFAS dehydration (loss of hydrogen bonds) is identified as one important factor for PFAS rejection.

The ML model interpretation plays a pivotal role in unraveling the complex mechanisms behind predictive analytics. In this study, ML model interpretation methods, including the SHAP method for the XGBoost model and heat maps highlighting the attention scores for the multimodal transformer model, unveil the underlying mechanisms of PFAS transport, based on which the performance predictions of ML models are determined. On one hand, the SHAP method for interpreting the XGBoost model reveals the general and macroscopic insights on how numerical values representing material properties and operational conditions govern the efficiency of NF and RO in removing PFASs. On the other hand, the multimodal transformer model integrated with molecular structures using SMILES strings provides a microscopic, molecular-level understanding of the detailed influences by chemical features of PFASs. This dual approach not only enhances the trust of the ML models but also promotes mechanistic knowledge of PFAS transport across polyamide membranes. The application of ML model interpretation methods, as demonstrated by the current study, can transform ML algorithms from simple predictive models into powerful tools of understanding and discovering complex governing factors behind the performance of membrane separation.

Furthermore, the results of this study, which identify key chemical features governing PFAS transport, have the potential to inform rational feature selection for membrane fabrication and optimization. For example, the highest importance of charge product and charged functional groups, as discerned by the interpretation of XGBoost and transformer models respectively, indicate that controlling electrostatic interactions between PFASs and membrane should be a prioritized strategy of promoting membrane selectivity to charged PFASs. Such knowledge attained by the ML models aligns with the outcomes of MD simulations, which demonstrates that PFAS molecules with the same carbon number but different functional groups experience varied energy barriers when traversing through polyamide membrane. Enhancing the electrostatic interactions by surface modification or the use of new monomers in interfacial polymerization is a promising direction of fabricating novel membrane materials with improved efficiency of PFAS removal. For instance, grafting materials with strong negative charge[66] or using monomers carrying negatively charged moieties in interfacial polymerization[67] has the potential of increasing electrostatic repulsion between PFAS and the membrane surface. Additionally, using monomers with lower dipole moments (e.g., melamine and pentaerythritol) may reduce the influence of electrostatic

attraction from dipole interactions. We acknowledge that the majority of PFASs in our database are anionic species in nature. This is largely due to the lack of commercial standards for cationic, zwitterionic, and most non-ionic PFASs, and thus current mass-spectrometry based methods for PFAS quantification can only target anionic PFASs in most cases. Nevertheless, the ML methods presented in this study can be employed beyond anionic PFASs if their quantifications and corresponding removal efficiencies by polyamide NF and RO membranes are available in the future. Also, the high attention scores of ether group in determining model prediction suggest the importance of functional group substitution in PFAS transport across polyamide NF and RO membranes, which is also supported by the results of MD simulations. This highlights research needs of evaluating membrane performance for removing diverse PFASs species other than PFCAs and PFSAs. In addition, with proper feature engineering to capture the distinct physical and chemical properties, the approaches developed in this work have the potential to predict the performance and gain mechanistic knowledge for other materials, such as novel membranes made of two-dimensional materials and nanochannels, as well as for the development of novel adsorbents and catalysts. Such applications enable the design of materials by leveraging an enhanced understanding of the intricate relationship between structural details and performance of functional materials.

Methods

Data collection

The data regarding PFAS rejection efficiencies of polyamide NF and RO membranes were collected from both the literature and our own experiments. For the data from the literature (Supplementary Data 1), which were used for ML model training, we extracted 457 data points either directly from tables presenting PFAS removal rates or by employing the Engauge Digitizer software to extract data from graphs. We also conducted experiments to evaluate the rejections of various PFASs from AFFF-contaminated feedwater using three commercial NF membranes and two modified NF membranes (Supplementary Table 3). From these experiments, 77 data points were acquired and used to assess the accuracy of ML models (Supplementary Data 1). The detailed membrane modification methods and the experimental procedure are described in the following sections.

Materials

The AFFF sample was obtained from 3 M Co. (St. Paul, Minnesota, US), and the standard solution (mass-labelled PFAS extraction, MPFAC-24ES) for PFASs quantification was purchased from Wellington Laboratories Inc. (Ontario, Canada). Methanol (> 99.9%, HPLC grade) was acquired from Honeywell (Muskegon, MI). Sodium chloride (NaCl, ≥99.0%), 2-propanol (IPA, ≥99.5%), ethanol (≥99%), hydrochloric acid (HCl, 36%), sodium hydroxide (NaOH, ≥97.0%), ammonium hydroxide (NH_3OH, 28.0–30.0%), acrylic acid ($CH_2CHCOOH$, ≥99.0%), potassium persulfate ($K_2S_2O_8$, 99.0%), sodium bisulfite ($Na_2S_2O_5$, ≥58.5%), dopamine chloride, tri(hydromethyl) aminomethane (Tris, ≥99.8%), and poly(ethylenimine) (PEI, MW 800) were all purchased from Sigma-Aldrich (St. Louis, MO). Deionized (DI) water (> 18 MΩ•cm) was provided from a water purification system (Millipore, Burlington, MA). Commercial flat sheet NF membranes (NF90, NF270, and NF) were purchased from DuPont FILMTEC (Wilmington, DE).

Membrane surface modification

Membrane surface modification was performed on NF270 membrane to achieve different membrane surface properties. We selected NF270 membrane for surface modification due to its loose structure and expected low PFAS removal efficiency (so that changing membrane surface properties has the best potential to alter the PFAS removal rates). Before membrane surface modification, NF270 membranes were initially immersed in a 25 v/v% IPA aqueous solution for 30 min and then washed thoroughly with DI water to remove impurities, followed by storage in DI water at 4 °C for further use. Two modification methods were employed to graft poly(acrylic acid) (PAA, a hydrophilic and negatively charged polymer) and poly(ethylenimine) (PEI, a hydrophilic and positively charged polymer) on the membrane surface. For the modification with PAA, a redox radical initiation reaction was employed, following the method reported by Belfer et al.[68]. Briefly, 1 M acrylic acid, 0.01 M $K_2S_2O_8$ and 0.01 M $Na_2S_2O_5$ solutions were prepared and poured onto the membrane surface to initiate the polymerization, which lasted for 1 h at the room temperature. The resulting PAA-modified membranes were then rinsed with DI water. For the modification with PEI, a dopamine-assisted grafting method was applied[69]. In short, 0.16 mM dopamine chloride was added to a Tris buffer solution (10 mM, pH 8.5) and then poured onto the membrane surface. After 1 h of dopamine polymerization reaction, polydopamine (PDA)-deposited membranes were rinsed with DI water, followed by immersion in 10 g L^-1 of PEI aqueous solution for 4 h at room temperature. Finally, PEI-modified membranes were rinsed with DI water. After the modification steps, the PAA- and PEI-modified membranes were stored at 4 °C until use.

Membrane characterization

Several techniques were employed to characterize the surface properties of the commercial and modified membranes. Water contact angles of the membranes were measured with a contact angle goniometer (Krüss DSA 10, Hamburg, Germany) using the sessile drop method. An electrokinetic analyzer with an adjustable gap cell (SurPASS, Anton-Paar, Graz, Austria) was employed to measure the zeta-potential of the membrane surface. The zeta potential measurements were performed with 1 mM KCl as background electrolytes in the pH range of 2 to 10, which was adjusted by titration of 0.1 M HCl and 0.1 M NaOH solutions. Detailed characteristics of the membranes can be found in Supplementary Table 3 and Supplementary Fig. 11.

The evaluation of PFAS rejections by NF membranes

The evaluation of PFAS rejections by the commercial and modified NF membranes was conducted in a lab-scale crossflow filtration system with an effective membrane area of 26.2 cm^2. The crossflow velocity was 21.3 cm s^-1 and the operating temperature was controlled at 25 ± 1 °C. Before the tests, the filtration system was thoroughly cleaned with 1% NH_4OH in DI water and ethanol (1:1) solution. The membranes were compacted overnight with DI water, and the pure water permeate flux was then measured by a digital flowmeter (FlowCal 5000, Tovatech, Plano, TX) to determine the water permeability coefficients of the membranes. Subsequently, NaCl was added to achieve an ionic strength of 10 mM in the feedwater for the measurement of NaCl rejection, followed by the addition of AFFF into the feed reservoir (the sum of detected PFASs concentrations was ~157 μg L^-1) with care to avoid bubble formation. The pH of the feedwater was adjusted (to pH of 5, 7, or 9) using 1 M HCl and 1 M NaOH solutions. The feed and permeate samples were collected in 15 mL polypropylene tubes (Corning Inc., Corning, NY) after 1 h operation and then stored at 4 °C until PFAS quantification.

PFAS quantification

For sample preparation, 100 μL of the collected feed or permeate sample was mixed with 100 μL of internal standard solution (20 μg L^-1, MPFAC-24ES in methanol) in a polypropylene vial (VWR, Radnor, PA). The prepared samples were analyzed by an Agilent 1290 liquid chromatography paired with an Agilent 6460 triple quadrupole mass spectrometer (LC-QQQ-MS, Agilent Technologies, Santa Clara, CA). The obtained concentrations were used for the calculation of PFAS rejection rates, which were used to evaluate the predictive performance of the ML models.

Molecular properties of PFASs

The SMILES strings of the PFASs included in our data set were used to build the molecular structure in Avogadro software (version 1.2.0). Such structures were imported into Marvin View (version 22.22.0, Chemaxon, Budapest, Hungary) for calculating the values of pK_a, maximum/minimum projection radius of PFAS molecule, and log K_{ow}. The computed values of the calculated molecular properties are listed in Supplementary Table 2 and Supplementary Data 1, with the obtained values being used as input variables for ML models. Additionally, atomic charges of the PFAS molecules were determined by the ORCA (version 5.0.3) software[70], and the dissociation of PFASs was considered when calculating the atomic charge of each atom. The atomic charge calculations were performed using the density functional theory method of B3LYP (a hybrid functional in density functional theory, which combines Becke's three-parameter exchange and Lee-Yang-Parr correlation) with def2-TZVPP basis set[71]. The impact of the solvent (i.e., water) was computed by utilizing the Conductor-like Polarizable Continuum Model (CPCM), and the values of static dielectric constant and refractive index of water are 80.4 and 1.3, respectively[72]. Finally, JANPA, an open-source JAVA software, was used for natural population analysis, which provides the value of atomic charge[73]. RDKit, a Python library, was then used to visualize the atomic charge of each atom in Supplementary Figs. 4–6.

Variable selection for ML models

We selected a set of variables pertaining to the properties of membranes and PFAS by considering membrane separation mechanisms such as size exclusion, Donnan exclusion, and adsorption[36]. As a result, eight variables were selected including compound size, log K_{ow}, charge product (a product of membrane charge index and compound charge), water contact angle (WCA) and MWCO of the membranes, hydraulic pressure, C_{in}, and ionic strength (IS) of the feedwater. Especially, compound size and charge product were calculated using the following equations[36].

$$\text{Compound size} = (\text{maximum projection radius} \times \text{minimum projection radius})^{0.5} \quad (1)$$

$$\text{Charge product} = \text{membrane charge index} \times \text{compound charge} \quad (2)$$

$$\text{Membrane charge index} = \text{isoelectric point} - \text{pH} \quad (3)$$

where the compound charge was calculated by weighting charged and neutral forms of PFAS at the experimental pH according to their respective proportions[36]; the isoelectric points of the membranes were determined by identifying the pH at which the zeta potentials of membranes became zero. For the multimodal transformer model, SMILES strings were tokenized, a process in which the strings are converted into numerical values, and these numerical values were incorporated as additional input variables.

ML models and their interpretation

We implemented two ML models, namely the XGBoost and multimodal transformer models, to predict PFAS removal by polyamide NF and RO membranes. The selection of the XGBoost model was due to its strong performance in many data science competitions and our previous publications[36,37,74]. While XGBoost has shown high predictive performance for structured and tabular data through its ensemble of decision trees[74], the multimodal transformer model leverages attention mechanisms, which enable efficient capturing of interdependencies especially in sequences like SMILES strings[46]. Considering the potentially important roles of chemical structure on PFAS transport across membranes, the multimodal transformer model that incorporates SMILES strings representing structural information has the potential to unveil the impacts of chemical features on PFAS removal by polyamide NF and RO membranes. To train the XGBoost and multimodal transformer models, we used 5-fold cross-validation and Bayesian optimization for hyperparameter tuning to avoid data leakage. Data were grouped according to experimental settings and combined with the corresponding PFAS names to create unique labels. Data sharing the same label were assigned to the same fold. This method ensured that data with similar experimental conditions (e.g., identical membrane type and PFAS, differing only in one variable) were kept within the same fold, effectively minimizing the risk of data leakage[36,37]. After optimizing the hyperparameters, the ML models were trained using the entire training dataset with the optimal hyper-parameters. The models' predictive capabilities were then evaluated using our own experimental data on PFAS rejection. The performance of the models was evaluated by using RMSE and MAE. These metrics are mathematically expressed as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y-\hat{y})^2} \quad (4)$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y-\hat{y}| \quad (5)$$

where n is the number of samples, y represents the true values (i.e., the target labels of the dataset), $\hat{y}$ represents the estimated values by the model.

After training the model, we employed SHAP, a commonly used XAI technique[36–38,51], to interpret the XGBoost model. In the SHAP method, the contribution of each variable to the model prediction is calculated by considering all possible combinations of variables and their corresponding contributions[51]. Thus, the use of SHAP enabled us to unveil the key factors governing PFAS removal by polyamide NF and RO membranes. For the multimodal transformer model, SMILES strings were utilized as additional inputs during model training. The SMILES strings were tokenized into numerical values, with each numerical value representing a specific chemical element, to facilitate their incorporation into the model. To ensure uniformity, the length of the tokens was standardized based on the longest set of tokens observed in the PFAS compounds present within the dataset. If the length of tokens was shorter than the maximum length, zeros were padded to achieve the required length. During the training of multimodal transformer model, attention scores were extracted from each input variable across different layers of multiple attention heads. These scores were then averaged for each atom within the structure of PFAS compounds to create heat maps, which visually represent the importance of each atom in determining PFAS removal by polyamide NF and RO membranes.

MD simulations of PFAS transport across polyamide membrane

Membrane preparation. Polyamide membrane composed of PIP and TMC was prepared with dimensions of $4.7 \times 4.7 \times 9\ nm^3$. The PIP-TMC polyamide membrane was chosen as the representative NF membrane that is typically used in experiments. Compared to polyamide membranes fabricated using m-phenylene diamine (MPD) and TMC, the looser PIP-TMC polyamide membrane is also more likely to unveil the relation between the functional group and the rejection of PFAS[25,75–77]. The cross-linked membranes are prepared following our previous work[78]. The monomers, PIP and TMC, are randomly placed inside a simulation box with PIP:TMC molar ratio of 1.5 and a target density of 1.3 g cc^-1. The system of monomers is energy-minimized and simulated crosslinking is initiated under a canonical ensemble at 300 K. At each cross-linking step, the reactive sites of PIP and TMC were

identified, and sites within specified cutoff distances of 0.45 nm were pulled towards each other using steered MD simulations (using virtual elastic springs). Once the reactive sites reached within 0.15 nm, physical amide bonds were assigned, appropriate H and Cl atoms were removed, and the topology of the molecule was updated. The crosslinking was continued until a cross-linking degree of 90% was achieved, which (excluding monomers) produced an O/N ratio of 1.14. At any intermediate crosslinking step if no reactive groups were found within the cutoff of 0.45 nm, the cutoff distance was increased by 0.05 nm. The monomeric TMC and PIP were removed and any remaining chloride atoms in TMCs were replaced with hydroxyl groups. Finally, one of the carboxyl groups in TMC containing exactly one amide bond was deprotonated (Fig. 5a). The charge density of our polyamide membrane is consistent with the charge density of membranes at pH 7.3 studied by Liu et al.[79].

Potential of mean force. The crosslinked polyamide membrane was inserted at the center of a box of dimension $4.7 \times 4.7 \times 19$ nm^3, followed by the addition of PFAS at the position (2.35, 2.35, 0 nm) and solvation with TIP3P water. The Swissparam[80] forcefield was applied to the polyamide membrane and PFAS. The system was energy minimized using the steepest descent, followed by canonical (NVT) ensemble for 50 ps, and isothermal-isobaric (NPT) ensemble for 20 ns. The temperature and pressure of the simulations were maintained at 300 K and 1 bar using a stochastic velocity rescaling thermostat with a time constant of 0.1 ps and Berendsen barostat with compressibility of 4.5×10^{-5} in z-direction (normal to the polyamide membrane) and 0 in x- and y-directions with a time constant of 10 ps. The initial configurations for umbrella sampling were generated by steering the PFAS molecules across the polyamide membrane with a force constant of 100 kJ mol^-1 nm^-2 and a pulling rate of 0.001 nm ps^-1. The reaction coordinate for umbrella sampling was defined as the displacement vector along the z-axis between the center of mass of PFAS and the polyamide membrane. A total of 120 windows were chosen for the reaction coordinate between ±6 nm for PIP-TMC. An umbrella potential with a force constant of 500 kJ mol^-1 nm^-2 was applied to PFAS in each window and equilibrated using eight cycles of 2.5 ns simulated annealing between 300 and 600 K and NPT equilibration of 5 ns (using the same thermostats and barostat). Finally, umbrella sampling simulations were performed with a force constant of 200 kJ mol^-1 nm^-2 using the Nose-Hoover thermostat[81] and Perrinello-Rahman barostat[82] for 80 ns, until convergence in the PMF was achieved. Periodic boundary conditions were used in all directions. The PMF profiles were calculated with the weighted histogram analysis method (WHAM[64]) using 300 bins, a tolerance of 10^-8, and the periodic assumption. The integrated autocorrelation time was determined for each window to assign relative weights to the windows[83]. All PMF calculations and analyses were performed using GROMACS 2022[84].

Hydrogen bond analysis. The hydrogen bonds formed between PFAS-polyamide membrane and PFAS-water were calculated using the MDAnalysis Python package[85]. The oxygen, sulfur, and fluorine atoms were treated as acceptors and oxygen and nitrogen atoms with at least one bound hydrogen were treated as hydrogen bond donors. The cutoff distance between donor and acceptor atoms and the angle between donor-hydrogen-acceptor for hydrogen bonds were set at 0.35 nm and 150°, respectively. The total number of hydrogen bonds was determined as a function of the reaction coordinate from all umbrella sampling windows. The reaction coordinate is binned with a width of 0.05 nm and all data points in a bin are used to determine the average and standard deviation.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this study are available from Supplementary Data 1, Source Data, and Supplementary Information. Source data are provided with this paper.

Code availability

The ML models and MD simulations data generated in this study have been deposited in the Zenodo database under accession code https://zenodo.org/records/13924418 and https://zenodo.org/records/13978801. The ML and MD simulations data used in this study are available in the Zenodo database using the provided accession code[86,87]. The ML and MD simulation data generated in this study are provided in the Supplementary Information/Source Data file.

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Experiment design: T.T. and Y.C., data collection and membrane experiments: S.P., performed ML modeling: N.J., T.T., and Y.C., performed MD simulation: S.M., J.Z., and Y.L., writing and editing: N.J., S.P., S.M., J.Z., J.B., Y.L., T.T., and Y.C.

Competing interests

The authors declare no competing interest.

Additional information

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